An Improved Sliding Model Observer Sensorless Control for PMSM Based on Fuzzy Logic Controller and DSOGI-FLL

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***Abstract*— The conventional sliding mode observer (SMO) employed in the sensorless control for the permanent magnet synchronous machine (PMSM) has the problems of chattering phenomena and phase delay, which results in large errors between the estimated and actual values of the rotor speed and position. Thus, this article proposes an improved SMO based on fuzzy logic controller (FLC) and dual second-order generalized integrator-frequency-locked loop (DSOGI-FLL) structure. The FLC is used to adjust the parameters of the SMO to improve the adaptive ability of the system and reduce the chattering. At the same time, in order to reduce the adverse effects from low-pass filter (LPF), DSOGI-FLL is used to accurately extract the back electromotive force (back EMF) and effectively eliminate the high-order harmonic components. Finally, the proposed method is verified by simulation and experiment, and compared with a novel super-twisting algorithm-based SMO with a complex vector generalized integrator (STASMO-CVGI), the results show that the improved SMO has a better performance.**

***Index Terms*— Dual second-order generalized integrator-frequency-locked loop (DSOGI-FLL), fuzzy logic controller (FLC), permanent magnet synchronous machine (PMSM), sensorless control, sliding mode observer (SMO).**

I. Introduction

THE permanent magnet synchronous machine (PMSM) is widely used as drive motors for electric vehicles due to its advantages of high power density and high efficiency. In addition to the self-performances of the PMSM, the performance of the system is affected by the control method. At present, the control strategy of PMSM mostly adopts the vector control. However, the vector control usually needs to install a mechanical sensor on the rotor shaft to accurately obtain the speed and position of the rotor. This not only increases the manufacturing cost but also reduces the reliability of the whole system, especially when the system operates in a relatively complex environment. For this reason, the sensorless vector control system has become a research hot spot in the field of electric vehicle drive in recent years.

At present, the sensorless control technology for PMSM mainly includes two methods, which are the high-frequency signal injection method and the method based on the back electromotive force (back EMF). The high-frequency injection method utilizes the salient pole effect of the motor to distinguish the rotor position and speed by injecting a high-frequency signal into the motor, which is usually used when the motor runs at the standstill or low-speed conditions [1], [2]. However, in the range of medium and high speed, the PMSM sensorless control scheme mostly adopts the model-based method, and the widely used methods include the extended Kalman filter (EKF) method, the model reference adaptive system (MRAS) method, and the sliding mode observer (SMO) method [3], [4]. The EKF has good anti-interference ability and fast-tracking speed, but the drawbacks of the large calculation amount and the high-performance processer requirement limit its use. In addition, although the MRAS has the advantages of simple structure and small estimation error, this method highly depends on the accuracy of the motor model parameters.

Sliding mode variable structure control (VSC) is a type of VSC. It is actually a kind of nonlinear control and widely used in various fields, such as motor control, robots, and so on. In different application situations, its focus is different and its combination forms are changeable [5]. In PMSM control, when it is used to replace the speed controller, it is called a sliding mode controller, and when it is used to replace the position sensor, it is called SMO. Compared to other sensorless control algorithms, i.e., EKF and MRAS, the SMO is not sensitive to system parameters and also has simple structure and strong anti-interference ability; therefore, it is widely used [6].

Unfortunately, the SMO is not completely perfect. The inherent chattering problem and the back EMF with high-order harmonics easily degrade the estimation accuracy of the SMO [7]. Using the sigmoid function to replace the switching function is an effective way to reduce system chattering

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phenomenon [8]. Second, low-pass filter (LPF) is often used to suppress harmonics in the back EMF. However, the LPF will lead to the amplitude attenuation and the phase shift [9], which often need to add additional phase compensation. Then, a feed-forward phase compensator method was proposed in [10] to directly phase shift the estimated back EMF, reducing the error of position estimation. In some studies, LPF is abandoned and adaptive filtering algorithms are used. In [11] and [12], an adaptive filter based on gradient descent and an adaptive filter based on frequency-adaptive delay signal cancellation (FADSC) were proposed, respectively. Both of them can effectively eliminate specific harmonics and effectively improve the estimation accuracy. However, these methods require tuning a large number of parameters, which increases the complexity of the system.

Therefore, in order to improve the estimation accuracy of the rotor position and speed, this article proposes an improved sliding model observer sensorless control method based on fuzzy logic controller (FLC) and a second-order generalized integrator-frequency-locked loop (SOGI-FLL), which can effectively solve the chattering phenomena of the SMO and filter out the high-order harmonics in the estimated back EMF at the same time. The specific innovations and contributions of this article are as follows: 1) an FLC is introduced to construct a fuzzy SMO (FSMO), realizing the adaptive sliding mode gain and sigmoid function parameters to reduce the chattering phenomena of the system; and 2) a SOGI-FLL algorithm is employed to replace LPF to filter out high-order harmonics in the estimation back EMF, which not only can weaken the problems of signal amplitude attenuation and phase delay but also has simple structure and high applicability. This article is organized as follows. In Section II, the basic principles of conventional SMO sensorless control algorithm are briefly introduced, and the shortcomings of the conventional SMO are analyzed. In Section III, an optimization method based on FLC is proposed to adjust the sliding mode gain and sigmoid function parameters to realize the adaptation of SMO parameters; at the same time, the dual SOGI-FLL (DSOGI-FLL) structure is completed to filter the high-order frequency harmonics in the back EMF, instead of the LPF. In Sections IV and V, the proposed improved SMO sensorless control method is verified by simulations and experiments, respectively. Finally, some conclusions are drawn in Section VI.

II. Analysis of Conventional SMO

A. Mathematical Model of PMSM

For the modeling of PMSM, some nonideal factors are ignored to simplify the derivation, such as cross-coupling magnetic saturation, iron loss, and magnet eddy current loss. Consequently, the current of PMSM in the two-phase stationary reference frame can be expressed as follows [13]:

$$\frac{d}{dt}\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \frac{1}{L_d}\begin{bmatrix} -R_s & 0 \\ 0 & -R_s \end{bmatrix}\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} + \frac{1}{L_d}\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - \frac{1}{L_d}\begin{bmatrix} E_\alpha \\ E_\beta \end{bmatrix} \tag{1}$$

$$\begin{bmatrix} E_\alpha \\ E_\beta \end{bmatrix} = \omega_e \psi_f \begin{bmatrix} -\sin\theta_e \\ \cos\theta_e \end{bmatrix} \tag{2}$$

where u_α and u_β are the α-axis and β-axis stator voltages, respectively, i_α and i_β are the α-axis and β-axis stator currents, respectively, L_d and L_q represent the d-axis and q-axis stator inductances, respectively, R_s is the stator-winding resistance, ω_e is the electrical angular velocity of the rotor, θ_e is the rotor electrical position, ψ_f is the permanent magnet flux linkage, and E_α and E_β represent the back-EMFs of α- and β-axes, respectively.

B. SMO Design

It can be seen from (2) that the back-EMFs contain the information of the rotor position. Therefore, the rotor position information can be obtained by observing the back-EMFs.

Based on the sliding mode variable structure theory, the sliding surface is selected as follows:

$$S(x) = \hat{i}_s - i_s \tag{3}$$

where $\hat{i}_s$ is the estimated current value and "^" means the estimation.

According to (1) and (3), the SMO is designed by the following equation:

$$\frac{d}{dt}\begin{bmatrix} \hat{i}_\alpha \\ \hat{i}_\beta \end{bmatrix} = \frac{1}{L_d}\begin{bmatrix} -R_s & 0 \\ 0 & -R_s \end{bmatrix}\begin{bmatrix} \hat{i}_\alpha \\ \hat{i}_\beta \end{bmatrix} + \frac{1}{L_d}\begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} - \frac{k}{L_d}\begin{bmatrix} \operatorname{sign}(S_\alpha(x)) \\ \operatorname{sign}(S_\beta(x)) \end{bmatrix} \tag{4}$$

where k is the gain of SMO. If the value of k is large enough, the SMO is able to be asymptotically stable [14].

However, the discontinuity of the sign function will make the estimated back EMF chattering violently. In order to reduce the chattering phenomenon, the sigmoid function of (5) is used instead of the sign function [15]

$$\operatorname{sigmoid}(S(x)) = \frac{2}{1+e^{-aS(x)}} - 1 \tag{5}$$

where a is a positive constant that adjusts the slope of the sigmoid function.

The stator current error equation can be obtained by substituting (5) into (4) and subtracting (1) as follows:

$$\frac{d}{dt}\begin{bmatrix} \bar{i}_\alpha \\ \bar{i}_\beta \end{bmatrix} = \begin{bmatrix} -\frac{R_s}{L_d} & 0 \\ 0 & -\frac{R_s}{L_d} \end{bmatrix}\begin{bmatrix} \bar{i}_\alpha \\ \bar{i}_\beta \end{bmatrix} + \frac{1}{L_d}\begin{bmatrix} E_\alpha - k\operatorname{sigmoid}(S_\alpha(x)) \\ E_\beta - k\operatorname{sigmoid}(S_\beta(x)) \end{bmatrix} \tag{6}$$

where $\bar{i}_\alpha = \hat{i}_\alpha - i_\alpha$ and $\bar{i}_\beta = \hat{i}_\beta - i_\beta$ are the current errors.

As can be seen from (6), it is only necessary to sample the PMSM current and compare it with the estimated current of the system; when their error converges to zero, the back EMF can be obtained

$$\begin{bmatrix} \hat{E}_\alpha \\ \hat{E}_\beta \end{bmatrix} = \begin{bmatrix} E_\alpha \\ E_\beta \end{bmatrix} = \begin{bmatrix} k\operatorname{sigmoid}(S_\alpha(x)) \\ k\operatorname{sigmoid}(S_\beta(x)) \end{bmatrix}. \tag{7}$$

Due to the dead zone of the inverter, the stator current is distorted, resulting in the presence of $6k \pm 1$-order harmonics in the back EMF [16], which must be filtered by LPF. The LPF will cause delay; hence, it must be compensated [17].

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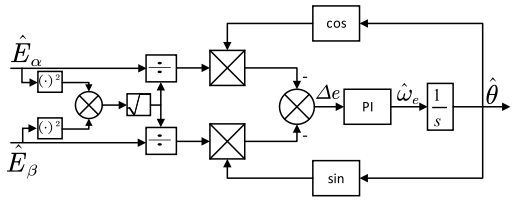


Fig. 1. Block diagram of Q-PLL.

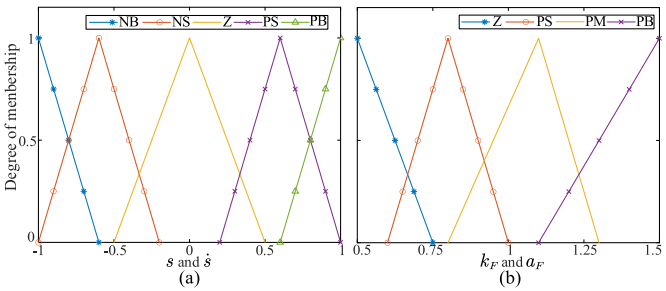


Fig. 2. Degree of membership function. (a) Input. (b) Output.

Usually, after filtering out the high-order harmonics, the rotor position can be calculated by the following equation:

$$\hat{\theta}_{\mathrm{e}} = -\arctan\left(\frac{\hat{E}_\alpha}{\hat{E}_\beta}\right). \tag{8}$$

However, due to the inherent chattering phenomena of the SMO and the division of the arctangent function, when the estimated back EMF crosses to zero, it is easy to introduce high-order harmonics into the estimation result, which makes the observation error larger. Therefore, in order to improve the accuracy of position estimation, a normalized quadrature phase-locked loop (Q-PLL) is usually used for processing the estimated back EMF [18]. Its structure is shown in Fig. 1.

C. Shortcomings of the Conventional SMO

As can be seen from (6), SMO has two parameters k and a to be tuned. The back EMF of the PMSM increases with the speed; in order to ensure the stability of the system, the value of k must be large enough. However, a larger value of k will bring more chattering, especially when the speed is low, which eventually lead to a decrease in the accuracy of the SMO estimation result.

The value of a is also a critical parameter, when the value of a is larger, the slope of the sigmoid function is larger, and the convergence speed is faster, but larger chattering will occur. Conversely, a small slope can reduce chattering in estimated values, but the speed of the system reaching the sliding surface will become slow. Therefore, in order to make the SMO has faster convergence and less chattering, an appropriate value of a needs to be chosen.

However, it is difficult to reduce the response time and chattering at the same time by adjusting the values of k and a. In addition, in order to filter out the high-order harmonics in the estimated back EMF, the convention SMO usually adopts the LPF with a fixed cutoff frequency. However, the LPF will bring delay to the system, resulting in the rotor speed variation. Although real-time compensation can improve the accuracy of the estimated position, it will reduce the rapid response of the system [19]. At the same time, the fixed cutoff frequency limits the speed range of motor as well, causing the variation range of the speed small.

Aiming at overcoming the abovementioned shortcomings of the SMO, this article proposes an improved SMO that will be introduced in detail in Section III.

III. Improvement of SMO

According to the previous analysis, the fixed sliding gain k and the parameters a of sigmoid function will lead to the limited observation range and the low observation accuracy of the SMO. The ideal value of k and the ideal value of a should be adaptive. When the system is far from the sliding surface, a larger value of both k and a should be selected to increase the speed of the system approaching the sliding surface and reduce the response time. When the system is close to the sliding surface, both the values of k and a should decrease, which will make the system chattering reduce. Meanwhile, the method of filtering the high-order harmonics of the estimated back EMF with the LPF should be avoided. Therefore, the improvement measures, namely, the FLC and DSOGI-FLL, will be proposed for solving these problems in the following.

A. Implementation of the FLC

FLC is an intelligent control algorithm based on expert experience. It does not require an accurate mathematical model of the controlled object, so it is very suitable for nonlinear and strongly coupled systems. Therefore, the combination of FLC and SMO can enhance the adaptive ability and robustness of the system [20], [21], [22].

A typical FLC mainly includes four parts: fuzzification, rules base, inference mechanism, and defuzzification. In this article, the FLC is designed as a system with two inputs and two outputs. The first input is the error between the estimated current and the actual current, named as s, and the second input is the derivative of the current error, called as $\dot{s}$.

First, the fuzzy set of input variables s and $\dot{s}$ is defined as {NB, NS, Z, PS, PB}, and the corresponding domain of discussion is arranged as $\{-1, -0.5, 0, 0.5, 1\}$, and the triangular membership function is used for fuzzification. In the same way, the fuzzy set of the output variables a_F and k_F is defined as {Z, PS, PM, PB}, and the corresponding domain range is [0, 1]. The membership functions are shown in Fig. 2, where N = negative, P = positive, B = big, M = middle, S = small, and Z = zero. Then, the fuzzy rules are defined. Fuzzy rules are a crucial part of the FLC, which determines the final control effect of the fuzzy controller. Based on theoretical analysis and experience, the basic rules are listed in Tables I and II. At the same time, the Mamdani method [11] is used for fuzzy inference, and the center of gravity method is used for defuzzification.

Finally, the two outputs k_F and a_F obtained by FLC are multiplied by the fixed sliding gain and the sigmoid function parameter, respectively; then, the adaptively changing sliding mode gain k and adaptive parameter a can be obtained. It is

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TABLE I
FUZZY RULE FOR A_F

a_F / s \ $\dot{s}$	NB	NS	Z	PS	PB
NB	PB	PB	PB	PM	PS
NS	PB	PM	PM	PS	PS
Z	PS	Z	Z	Z	PS
PS	PS	PS	PM	PB	PB
PB	PS	PM	PB	PB	PB

TABLE II
FUZZY RULE FOR K_F

k_F / s \ $\dot{s}$	NB	NS	Z	PS	PB
NB	PB	PB	PM	PS	Z
NS	PM	PM	PS	Z	PS
Z	PS	Z	Z	Z	PS
PS	PS	Z	PS	PM	PM
PB	Z	PM	PB	PB	PB

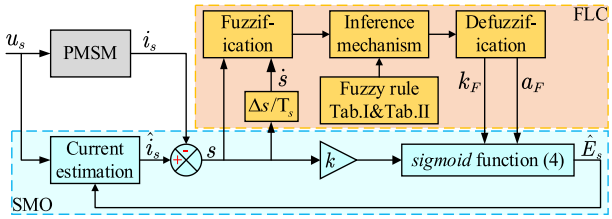


Fig. 3. Block diagram of FSMO.

worth noting that both adaptive parameters need to meet the requirements of system stability. Its system structure is shown in Fig. 3.

B. Improved Filtering by DSOGI-FLL

In order to improve the response speed and estimation accuracy of the system, this article introduces the SOGI-FLL algorithm, which is widely used in the extraction of power grid voltage synchronization signals, instead of LPF. The basic structure of SOGI-FLL mainly includes two parts: SOGI and FLL. The basic structure diagrams are shown in Figs. 4 and 5.

According to Fig. 4, the transfer function of SOGI can be obtained as follows:

$$D(s) = \frac{v_o}{v_{\text{in}}} = \frac{k_s\omega' s}{s^2 + k_s\omega' s + \omega'^2} \tag{9}$$

$$Q(s) = \frac{qv_o}{v_{\text{in}}} = \frac{k_s\omega'^2}{s^2 + k_s\omega' s + \omega'^2} \tag{10}$$

where v_{in} is the input signal, while v_o and qv_o are two in-quadrature output signals, ε_f is the frequency error signal, and ω' indicates the resonance frequency of the SOGI.

According to (9) and (10), $D(s)$ is a bandpass filter, and $Q(s)$ is an LPF, whose output signal qv_o lags v_o by 90°.

In order to facilitate the theoretical proof of the effectiveness of DSOGI-FLL, the estimated back EMF is expressed by the following equation:

$$\hat{e} = E_{\text{dc}} + E_1 \sin(\omega_1 t + \varphi_1) + \sum E_h \sin(\omega_h t + \varphi_h) \tag{11}$$

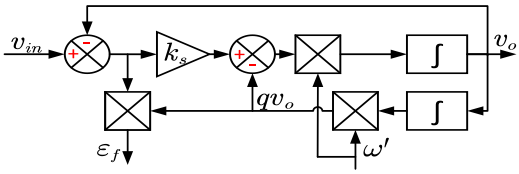


Fig. 4. Block diagram of SOGI.

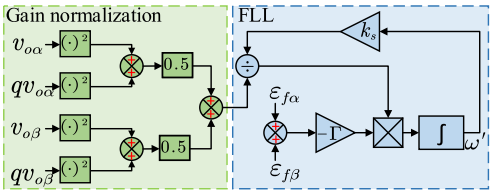


Fig. 5. FLL with a gain normalization unit.

where the first term is the dc component, the second term is the fundamental component, and the third term is the harmonic component.

The filtered estimated back EMF is expressed as follows:

$$\hat{E}_f(s) = \hat{E}(s)D(s). \tag{12}$$

For the dc component, the final value theorem can be deduced

$$\lim_{s\to 0} s\frac{E_{\text{dc}}}{s}D(s) = 0. \tag{13}$$

It shows that in the steady state, the dc component is zero.

For the fundamental component, it is assumed $\omega' = \omega$, let $s = j\omega'$, substitute into (9), then $D(j\omega') = 1e^{j}0°$. It is proved that before and after filtering, the amplitude and phase of the fundamental component remain unchanged.

For the harmonic components, let $s = jh\omega'$, it can be obtained

$$D(jh\omega) = \frac{1}{\sqrt{1 + \left(\frac{1-h^2}{k_s h}\right)^2}} e^{j\left(\pi - \arctan\frac{k_s h}{1-h^2}\right)} = A_h e^{j\gamma_h}. \tag{14}$$

After filtering, the harmonic components in the back EMF become

$$\hat{e}_h = \sum A_h E_h \sin(\omega_h t + \varphi_h + \gamma_h) \tag{15}$$

where A_h is the amplitude gain and γ_h is the phase offset.

Comprehensive (11)–(15), the estimated back EMF after filtering can be expressed as follows:

$$\hat{e}_f = E_1 \sin(\omega_1 t + \varphi_1) + \sum A_h E_h \sin(\omega_h t + \varphi_h + \gamma_h). \tag{16}$$

Assume that the fundamental frequency is 50 Hz, $k_s = (2)^{1/2}$. Since the harmonics in the back EMF are mainly the fifth and seventh harmonics [23], substituting $h = 5$ and $h = 7$ into (14), respectively, it can be obtained that the amplitude attenuation values of the fifth harmonic and the seventh harmonic are −11 and −14 dB, respectively. While using an LPF with a cutoff frequency of 500 Hz, the corresponding amplitude attenuations are −1 and −5 dB, respectively. It can be seen that using SOGI filtering can not only effectively filters out the harmonic components in the back EMF but also

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keeps the amplitude and phase of the fundamental component unchanged.

At the same time, according to the above derivation, it can be seen that ω' and k_s determine the bandwidth center frequency and bandwidth of the SOGI, respectively. The response speed of SOGI is positively relative to the value of k_s, but the filtering effect will be reduced. If the value of k_s is smaller, the corresponding bandwidth will become smaller and the signal selectivity will be better, but the response time is slow.

In order to take into account both response speed and filtering effect, $D(s)$ is analyzed. Obviously, $D(s)$ is a typical second-order system. On the basis of the second-order system setting principle, when the optimal damping ratio $\xi = 0.707$ is selected, the adjustment time and overshoot of the system are satisfactory. Therefore, it can be obtained $k_s = (2)^{1/2}$.

The signal of the specified frequency can be extracted from v_{in} by changing ω', so the FLL is added to realize the self-adaptation of SOGI to the frequency. Adding an integrator with negative gain $-\Gamma$ to the FLL allows ω' to approximate the actual frequency of the input signal. At the same time, the normalization of the output signal v_o can make the response frequency of FLL only depend on the $-\Gamma$, without depending on the signal frequency. Based on comprehensive simulation and experiment, when $\Gamma = 50$, the system has better results in balancing the adjustment speed and accuracy.

Since the estimated back EMF $\hat{E}_\alpha$ and $\hat{E}_\beta$ have the same frequency, it is possible to use only one FLL for frequency estimation and combine with two SOGIs to form a DSOGI-FLL. This structure not only reduces the complexity of the system but also eliminates the second harmonic in the signal, improving the accuracy and response speed of the estimated frequency [24].

C. Digital Implementation of SOGI

In order to implement SOGI in DSP, it needs to be digitized. For the digital implementation, different discretization methods will get different results, and the performance of the system will be affected. At present, the actually available discretization method for SOGI digitization is to discretize the integrator. The most common methods of obtaining a discrete-time integrator include the Tustin method, the forward Euler method, and the backward Euler method [25]. The two output signals of the digital SOGI obtained by the Tustin method can maintain good orthogonality, but there will be a large error when the signal frequency changes [26]. When using the Euler method, the amplitude–frequency characteristics and orthogonality of digital SOGI will be affected. Therefore, in order to maintain good performance of digitized SOGI, this article uses the Tustin with prewarping method to weaken the shortcoming of the common Tustin method, which the discrete-time integrator is equivalent to

$$\frac{\tan(\omega' T_s/2)}{\omega'}\frac{1+z^{-1}}{1-z^{-1}}. \tag{17}$$

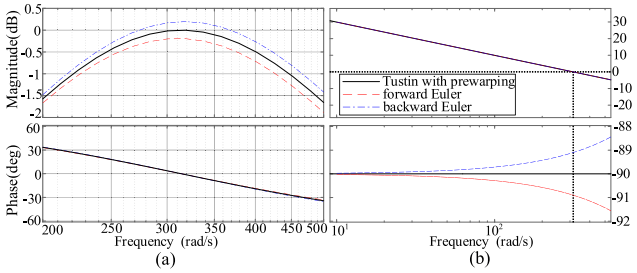


Fig. 6. Bode diagram of the digital SOGI. (a) v_o/v_{in} and (b) qv_o/v_o.

The discretized transfer functions of digital SOGI are given as follows:

$$D(z) = \frac{2k_s \sin(\omega' T_s)(1-z^{-2})}{k_s \sin(\omega' T_s)+2-4k_s \cos(\omega' T_s)z^{-1}+(2-k_s \sin(\omega' T_s))z^{-2}} \tag{18}$$

$$Q(z) = \frac{k_s(1-\cos(\omega' T_s))(1+z^{-1})^2}{k_s \sin(\omega' T_s)+2-4k_s \cos(\omega' T_s)z^{-1}+(2-k_s \sin(\omega' T_s))z^{-2}}. \tag{19}$$

When the system steady-state error is 0, the frequency of the output signal can be expressed as follows:

$$f' = \arccos\left(-\frac{\tan^2(\omega' T_s/2)-1}{\sec^2(\omega' T_s/2)}\right)\frac{1}{2\pi T_s}. \tag{20}$$

Assume $T_s = 100\ \mu$s and $\omega' = 100\ \pi$ and then substitute in (20), the result is $f' = 50$ Hz. Under the same conditions, the result of the Tustin method is $f' = 49.59$ Hz. Combined with Fig. 6, it can be seen that the digital SOGI obtained by Tustin with prewarping method maintains good performance.

IV. Simulation Results

In order to verify the effectiveness of the proposed method, this article conducts simulation and experimental comparisons with another improved SMO method, supertwisting algorithm-based SMO with a complex vector generalized integrator (STASMO-CVGI), in [27]. The system block diagram is established, as shown in Fig. 7. The parameters of PMSM control system are tabulated in Table III. Additionally, the relevant parameters for the SMO are set to $k = 1$ and $a = 4$. The gain parameters of the DSOGI-FLL are set to $k_s = (2)^{1/2}$ and $\Gamma = 50$. The back EMF is output in per unit, and the reference value is $311/(3)^{1/2}$.

A. Validation of the Effectiveness of the FSMO

The estimation performance of the back EMF is one of the important factors affecting the accuracy of the SMO. Consequently, the estimated back EMF results based on the FSMO should be evaluated first. Fig. 8 shows the comparison of the estimated back EMF by the STASMO-CVGI and the FSMO. Fig. 8(a) shows that the estimated back EMF contains

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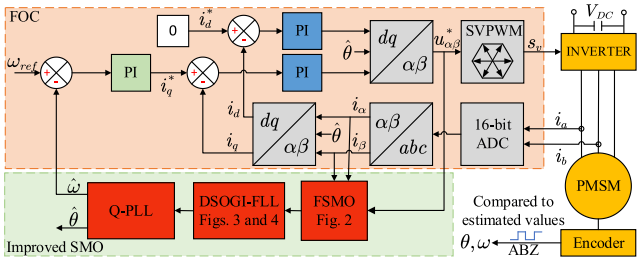


Fig. 7. Block of the improved SMO senseless control system of PMSM.

TABLE III
PARAMETERS OF PMSM

Parameters	Value	Unit
Rate power	2	kW
Rate line voltage	220	V
Rate speed	2000	rpm
Rotor inertia	0.00153	Kg·m^2
Rate load	7.7	N·m
Number of pole pairs	4	-
d-axis inductance	1.47	mH
q-axis inductance	1.47	mH
Staor resistance	0.505	Ω
Flux of permanent magnet	0.1326	Wb
Sampling frequency	10	kHz
Current loop control frequency	10	kHz
Speed loop control frequency	1	kHz

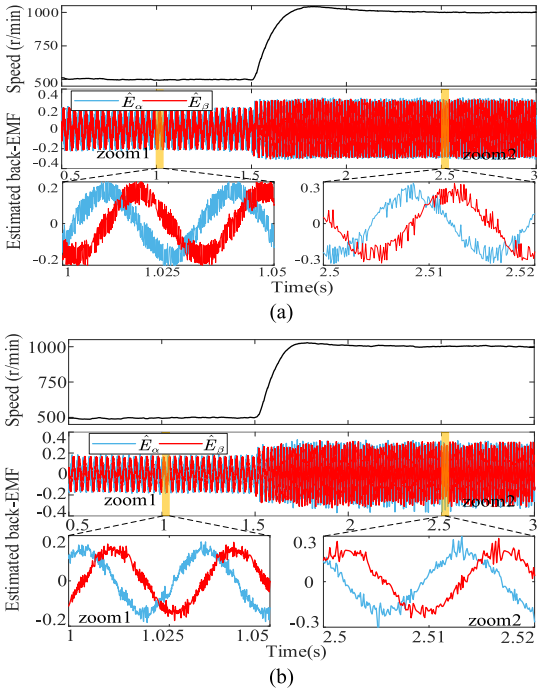


Fig. 8. Simulation results of the estimated back EMF. (a) STASMO. (b) FSMO.

a large number of harmonic components at low speed using STASMO-CVGI, while the harmonic content will decrease with the increase in speed. In contrast, since FSMO can adaptively adjust the system parameters, the harmonic content in the estimated back EMF is significantly smaller than the STASMO-CVGI and the waveform is smoother. The results demonstrate the effectiveness of FSMO.

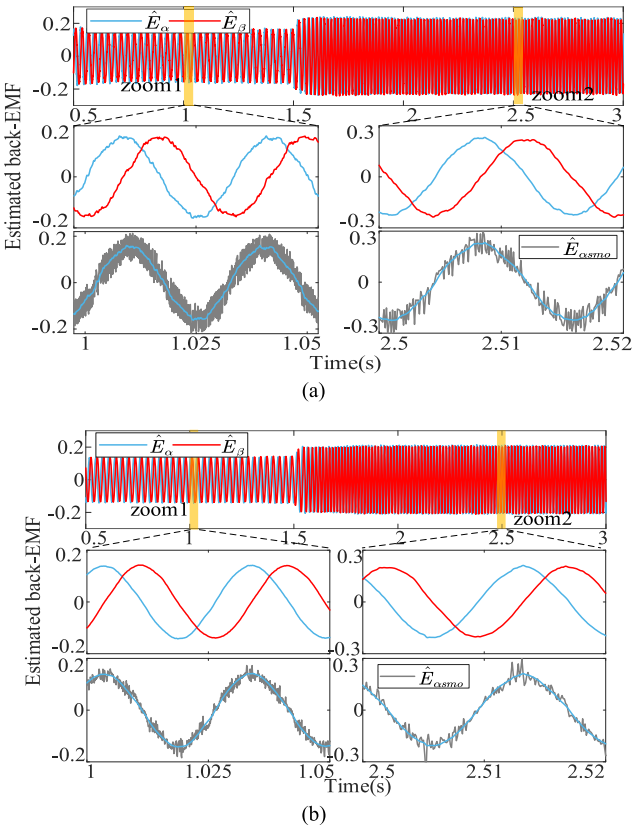


Fig. 9. Estimated back EMF without and with filter, acceleration from 500 to 1000 r/min. (a) CVGI. (b) DSOGI-FLL.

B. Validation of the Effectiveness of the DSOGI-FLL

In order to verify the effectiveness of DSOGI-FLL structure, first, let PMSM run at 500 r/min under no-load condition, increase the speed to 1000 r/min at 1.5 s, and the motor enters a steady state at 2 s.

The waveforms before and after filtering $\hat{E}_s$ are shown in Fig. 9. It can be seen that $\hat{E}_s$ is filtered by two kinds of filters, and both the waveforms become smoother and closer to the standard sine wave than the SMO direct output $\hat{E}_{\alpha\text{smo}}$. However, the waveform amplitude and phase outputted by the DSOGI-FLL are closer to $\hat{E}_{\alpha\text{smo}}$, especially when the speed is increased to 1000 r/min and enters the steady state. Although the method based on CVGI filtering does not have the same phase delay as LPF, its waveform is obviously not as smooth as DSOGI-FLL, and there are still significant harmonic components.

The fast Fourier transform analysis of $\hat{E}_\alpha$ and the Lissajous curves of the estimated back EMF after the CVGI and DSOGI-FLL filter are shown in Fig. 10, when PMSM is running at a steady state of 500 r/min. It can be seen that the back EMF outputted by the DSOGI-FLL filter has less harmonic content and the Lissajous curve is smoother and closer to a circle. The simulation results are consistent with the theoretically derived results.

C. Without Load Variable Speed Condition

Aiming at further verifying the performance of the improved SMO sensorless control proposed in this article, the following

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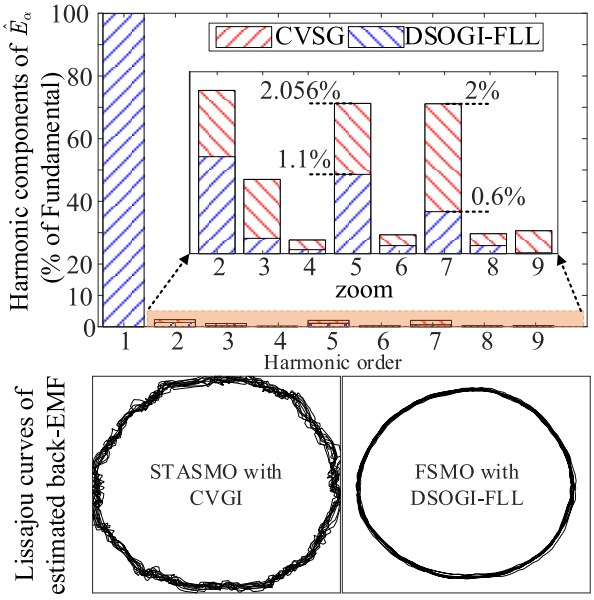


Fig. 10. FFT of the filtering $\hat{E}_\alpha$ and Lissajous curve of estimated back EMF.

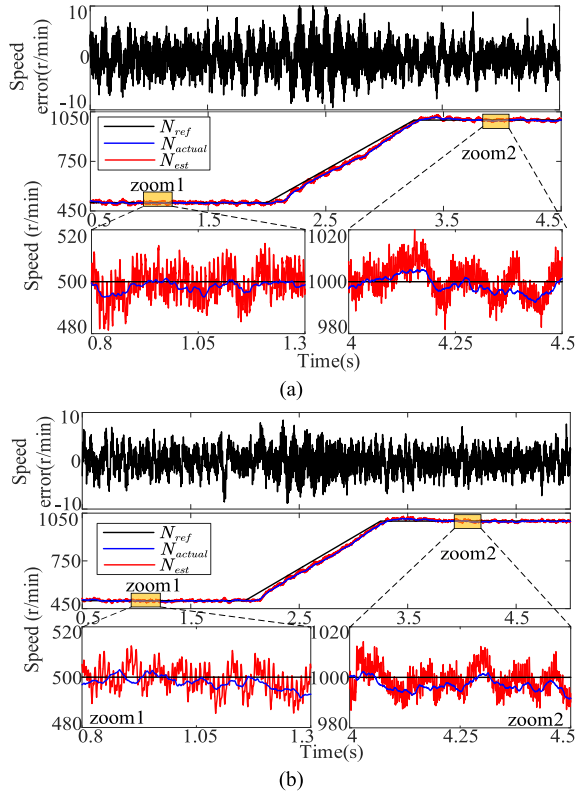


Fig. 11. Simulation comparison of speed control performance from 500 to 1000 r/min. (a) STASMO-CVGI. (b) Improved SMO.

simulations are implemented using the estimated speed and position.

PMSM starts up to the speed of 500 r/min and then increases the reference speed to 1000 r/min at 2 s. Figs. 11(a) and 12(a) are the estimated speed and estimated position of the rotor by the STASMO-CVGI. Figs. 11(b) and 12(b) are the estimated rotor speed and position obtained by the improved SMO.

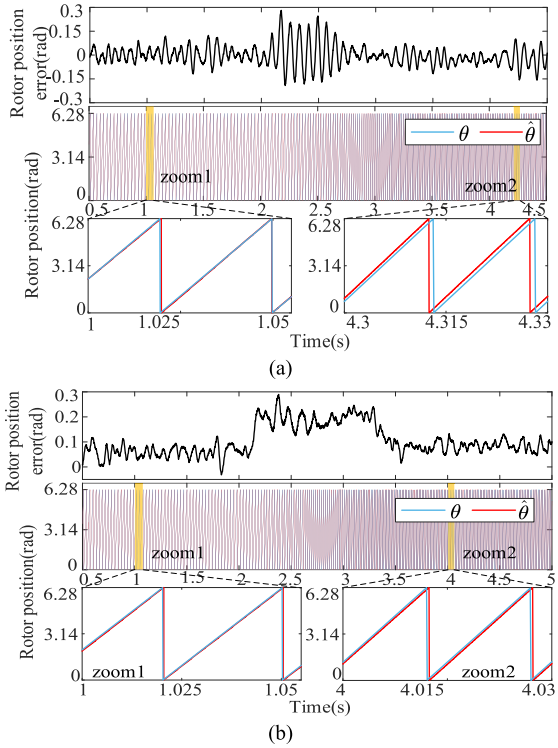


Fig. 12. Rotor position tracking performance comparison. (a) STASMO-CVGI. (b) Improved SMO.

It can be seen from Fig. 11 that the estimated speed waveforms of the STASMO-CVGI have obvious chattering and large error, and the maximum error is around 8 r/min. In addition, overshoot is occurred in the process of speed increase, and the maximum speed error reaches to about 10 r/min. After entering the steady state, the speed estimation error becomes larger. In comparison, the speed waveform chattering of the improved SMO is reduced, and the maximum error is about 5 r/min. When the speed changes, the error does not change significantly and the estimated speed basically coincides with the actual speed.

Fig. 12(a) shows that although STASMO-CVGI can reduce the chattering phenomenon, due to the fixed parameters, the error between the rotor position estimate and the actual position is constantly changing. At 500 r/min, the maximum position error is about 0.12 rad. When the speed is increased to 1000 r/min, the maximum position error becomes larger, about 0.15 rad. As shown in Fig. 12(b), the estimated position from the improved SMO has higher accuracy, and the error maintains at approximate 0.07 rad before and after the speed changes. In the dynamic process, although the tracking ability of DSOGI-FLL becomes worse and the maximum position error becomes 0.2 rad, it is still smaller than 0.3 rad of STASMO-CVGI.

D. Constant Speed With Load Disturbance

The anti-interference ability of the SMO is investigated as well. The speed of the PMSM is constant at 1000 r/min, the load suddenly increases to 50% rated load at 1.5 s, and the simulation results are shown in Fig. 13.

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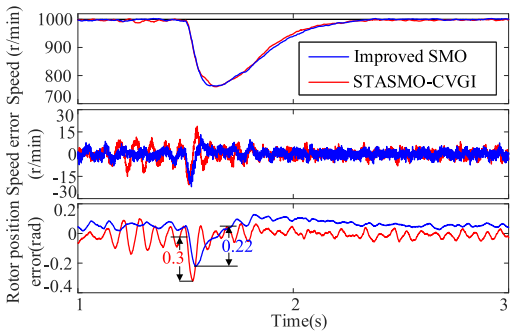


Fig. 13. Speed and position estimation errors under 1000 r/min with 50% rated load disturbance.

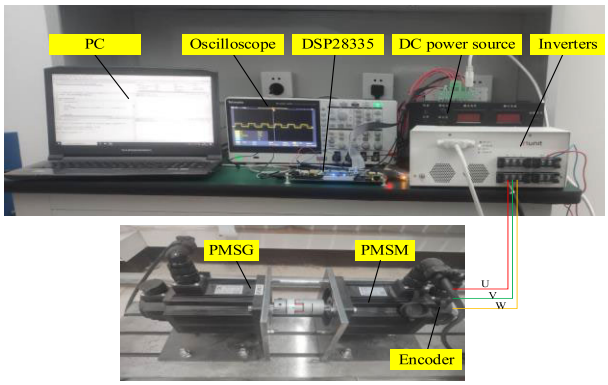


Fig. 14. Experimental platform.

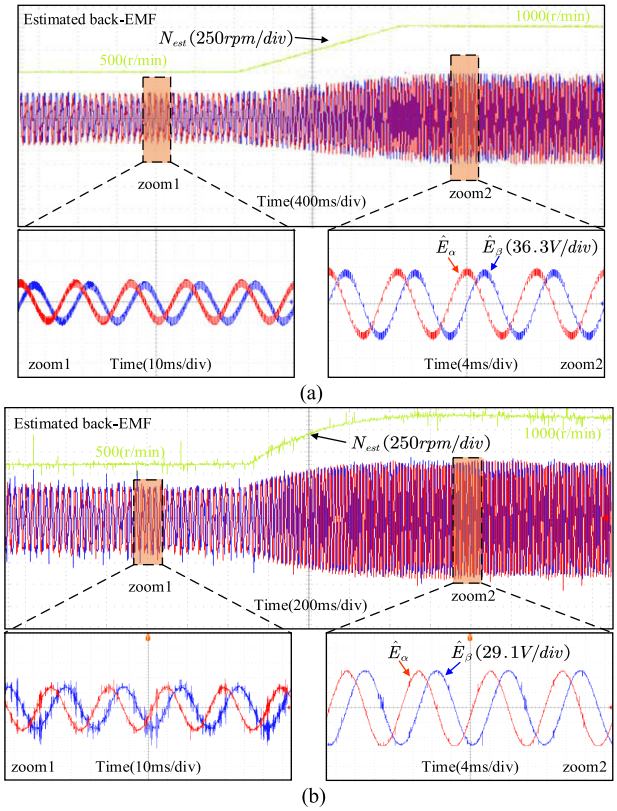


Fig. 15. Experimental results of the estimated back EMF. (a) STASMO. (b) FSMO.

It can be seen from Fig. 13 that when the system is disturbed by the load, both can respond quickly, and the maximum error of the speed in the dynamic process is 23 r/min. After entering the steady state, the average error of the STASMO-CVGI is 8 r/min, while the average error of the improved SMO becomes 5 r/min. At the same time, the maximum position errors of the STASMO-CVGI and the improved SMO during the dynamic process are 0.3 and 0.22 rad, respectively. The steady-state position error of the STASMO-CVGI changes between -0.15 and -0.12 rad. Conversely, the improved SMO has higher estimation accuracy under the action of FLC adjustment, and the steady-state position error is changed between 0.08 and 0.1 rad. Obviously, the improved SMO steady-state position error is slightly smaller and the anti-interference ability is stronger.

V. Experimental Results

In this section, the proposed sensorless control method is experimentally verified on the PMSM control platform shown in Fig. 14. The PMSM parameters of the experimental platform are shown in Table III. All data are observed through the oscilloscope. An incremental encoder is used to obtain the actual rotor position, which is convenient for comparison with the estimated rotor position. The control frequency and the switching frequency of the inverter are set to 10 kHz, and the dead time of the inverter is set as 2 μs.

A. Validation of the Effectiveness of the FLC

Fig. 15 compares the estimated back EMF waveforms using FSMO and STASMO for PMSM running at 500 and 1000 r/min, respectively. It can be seen that the back EMF waveform estimated by the FSMO is more sinusoidal. Especially, when the speed of the motor is low and the back EMF is small, the FSMO can effectively reduce the chattering phenomenon of the system. However, there are more harmonics in the estimated back EMF by STASMO, which is much more severe at low speed. From the comparison results, it can be seen that the adaptive gain of SMO is achieved with the support of FLC, which improves the performance of SMO and significantly reduces the chattering phenomenon.

B. Validation of the Effectiveness of the DSOGI-FLL

Fig. 16 compares the filtering performance of the estimated back EMF at 500 and 1000 r/min between the CVGI and DSOGI-FLL, respectively. It can be seen that the back EMF filtered by CVGI is slightly shifted, and this phenomenon is more obvious after the speed increases to 1000 r/min, but the filtering effect is still better than the LPF, the phase difference between the back EMF given by DSOGI-FLL and the α-axis estimated back EMF is tiny, and the curve amplitude basically has no attenuation. This proves that DSOGI-FLL has a better filtering effect than CVGI, with a smoother waveform and less harmonic content, which is beneficial to improve the estimation accuracy of the observer.

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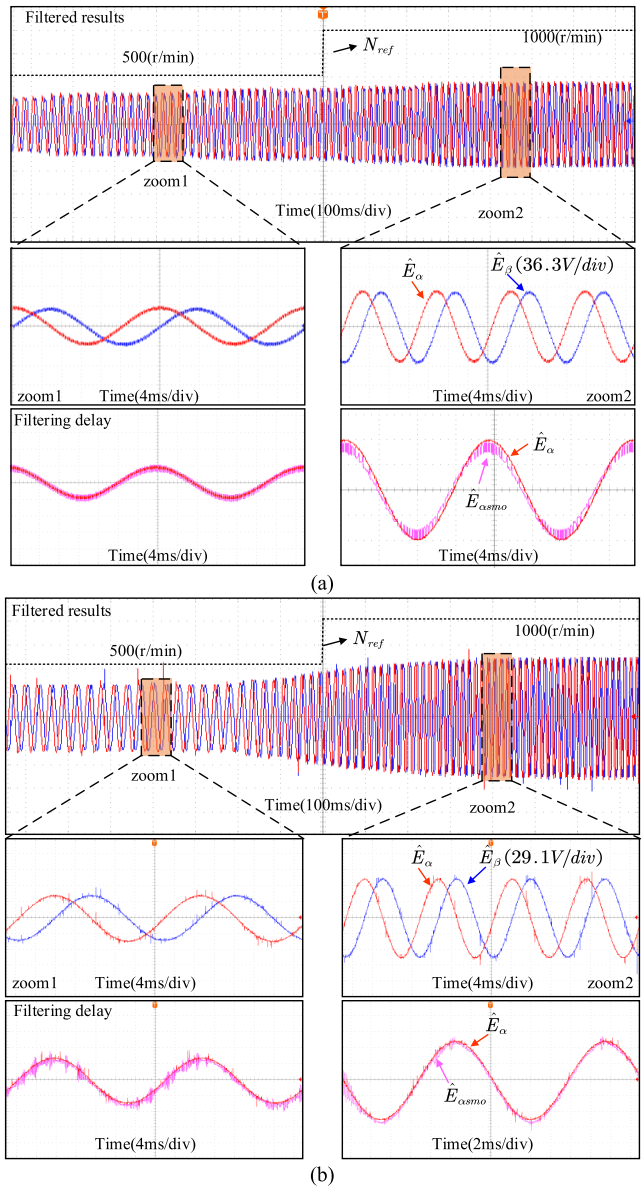


Fig. 16. Experimental results of the estimated back EMF filtered by different filters, acceleration from 500 to 1000 r/min. (a) CVGI. (b) DSOGI-FLL.

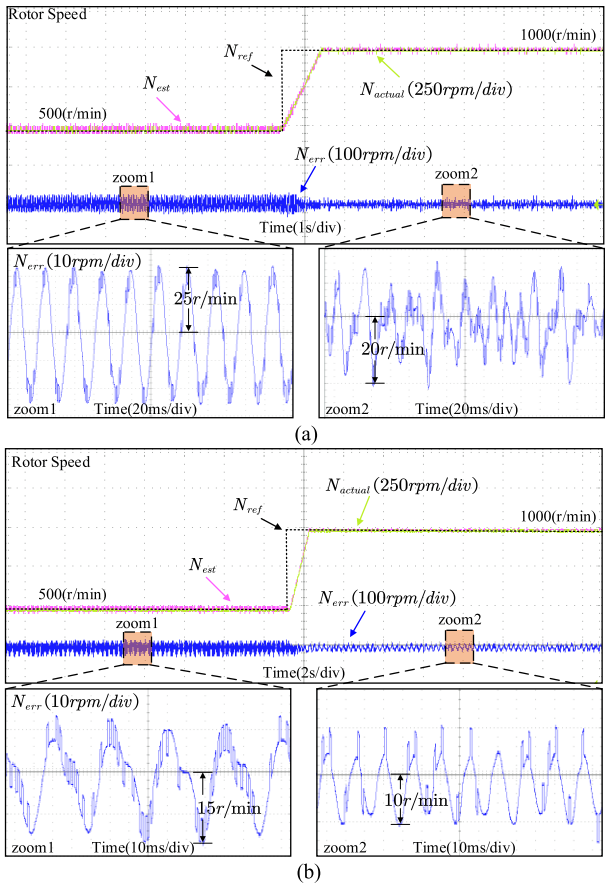


Fig. 17. Experimental results of speed estimation when the reference speed is from 500 to 1000 r/min. (a) STASMO-CVGI. (b) Improved SMO.

C. Without Load Variable Speed Condition

Figs. 17 and 18 show the comparative performances when the reference speed changes from 500 to 1000 r/min without load. Among them, Figs. 17(a) and 18(a) are the results of the STASMO-CVGI sensorless control strategy, and Figs. 17(b) and 18(b) are the results of the improved SMO sensorless control strategy proposed in this article. It can be seen from the experimental results that when the system is running at 500 r/min, the maximum error of rotor speed estimation of STASMO-CVGI is 25 r/min, and the rotor position estimation error is 0.32 rad. In comparison, the improved SMO has a maximum rotor speed estimation error about 15 r/min and a rotor position estimation error of 0.1256 rad at 500 r/min; the estimation accuracy is higher than STASMO-CVGI. As the increase of the rotor speed, the back EMF becomes larger, and the maximum error of the estimated rotor speed of the two sensorless control strategies has decreased. The maximum speed estimation error of the STASMO-CVGI is 20 r/min, while the maximum error of the improved SMO speed is 10 r/min. However, at 1000 r/min, due to the phase delay caused by CVGI and the fixed sliding mode coefficients, the maximum error of the rotor position estimation is still very large, about 0.2 rad. On the contrary, the improved SMO reduces the maximum error of rotor position estimation to 0.08 rad when the rotor speed increases. The result shows that in comparison with STASMO-CVGI, the estimation values of the improved SMO are more accurate and the performance is better, which proves the effectiveness and practicability of the improved SMO.

D. Constant Speed With Load Disturbance

Figs. 19 and 20 show the experimental results of STASMO-CVGI and the improved SMO at 1000 r/min with 20% rated load disturbance. During the load process, the estimation error of the improved SMO remains around 10 r/min. The maximum error of STASMO-CVGI is 20 r/min, and its steady-state error becomes larger after loading. Simultaneously, it can be seen that the rotor position estimation errors are 0.184 and 0.152 rad, respectively, and the error variation of the improved SMO is significantly smaller than the STASMO-CVGI. Meanwhile, the improved SMO has

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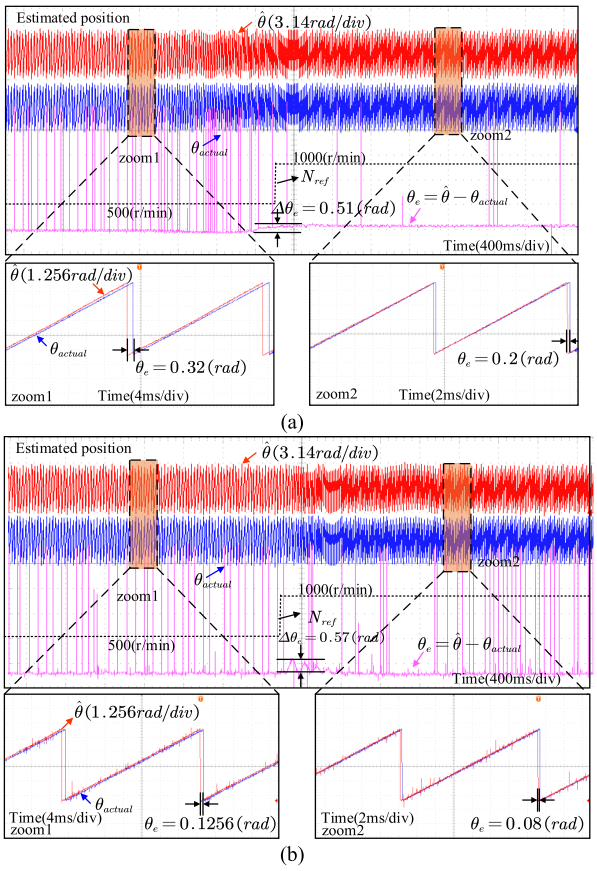


Fig. 18. Comparison of experimental results for rotor position estimation when the reference speed is from 500 to 1000 r/min. (a) STASMO-CVGI. (b) Improved SMO.

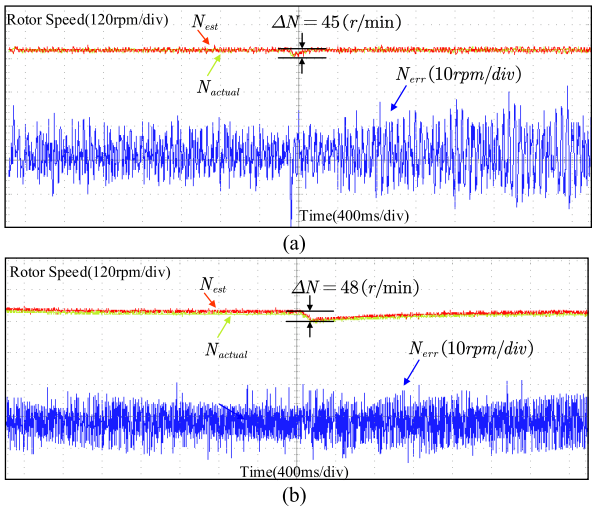


Fig. 19. Comparison of experimental results for speed estimation when the PMSM running at 1000 r/min with 20% rated load disturbance. (a) STASMO-CVGI. (b) Improved SMO.

better anti-interference ability and smaller fluctuations in stator current, as shown in Fig. 21.

The comparison of key performance index among different control methods is shown in Table IV under the condition of PMSM running at 1000 r/min. It can be found that all

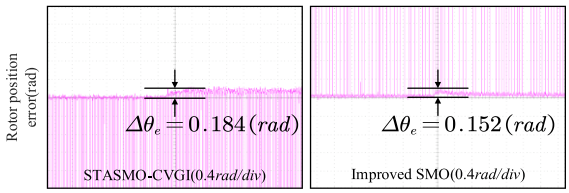


Fig. 20. Rotor position error when 20% rated load impacts at steady-state speed of 1000 r/min.

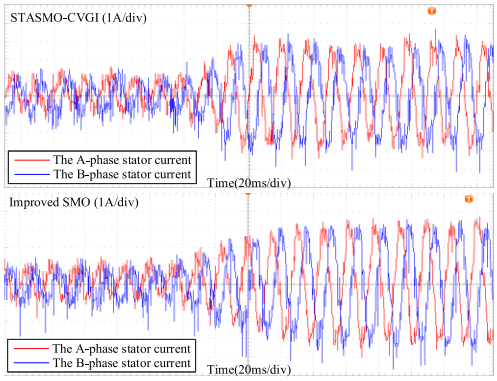


Fig. 21. Stator current waveforms when 20% rated load impacts at steady-state speed of 1000 r/min.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT SMO CONTROL METHODS

Estimation method	Harmonics components	Amplitude attenuation	Phase delay	θ_e rad	N_{err} r/min
SMO+LPF	High	Obvious	Obvious	0.32	30
STASMO+CVGI	High	None	Tiny	0.2	20
FSMO+DSOGI	Low	None	None	0.08	10

performance indicators of the improved SMO are better than the SMO + LPF and the STASMO + CVGI.

E. Limitation of the Proposed Method

1) The improved SMO is an estimation method based on the back EMF of PMSM. However, when the back EMF is not obvious, it will bring a certain control error; therefore, other control methods need to be used for start-up and low-speed range.

2) During the dynamic process, the estimated angle error becomes larger, which may affect the dynamic performance of the system.

VI. CONCLUSION

In this article, an improved SMO sensorless control method based on FLC and DSOGI-FLL structure is proposed. The research results show that the FLC can improve the adaptive ability of SMO, shorten the convergence time of the observer, and reduce the chattering and the estimation error. At the same time, the DSOGI-FLL has better filtering effect, showing that the estimated back EMF filtered by the DSOGI-FLL is basically consistent with the original signal in phase and amplitude. The experimental results verify that the improved SMO can not only track the position and speed of rotor quickly and accurately but also has the characteristics of fast response and strong robustness.

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REFERENCES

[1] Z. Wang, K. Yu, Y. Li, and M. Gu, "Position sensorless control of dual three-phase IPMSM drives with high-frequency square-wave voltage injection," *IEEE Trans. Ind. Electron.*, vol. 70, no. 10, pp. 9925–9934, Oct. 2023.

[2] S. Du, Q. Li, X. Zhu, L. Zhang, and Y. Zuo, "Fault-tolerant control of position sensor failure for PMSM at zero and low speed based on high frequency injection," *Proc. CSEE*, vol. 39, no. 10, pp. 3038–3047, May 2019.

[3] Y. Jiang, W. Xu, C. Mu, J. Zhu, and R. Dian, "An improved third-order generalized integral flux observer for sensorless drive of PMSMs," *IEEE Trans. Ind. Electron.*, vol. 66, no. 12, pp. 9149–9160, Dec. 2019.

[4] Z. Qiao, T. Shi, Y. Wang, Y. Yan, C. Xia, and X. He, "New sliding-mode observer for position sensorless control of permanent-magnet synchronous motor," *IEEE Trans. Ind. Electron.*, vol. 60, no. 2, pp. 710–719, Feb. 2013.

[5] Z. Li, F. Wang, D. Ke, J. Li, and W. Zhang, "Robust continuous model predictive speed and current control for PMSM with adaptive integral sliding-mode approach," *IEEE Trans. Power Electron.*, vol. 36, no. 12, pp. 14398–14408, Dec. 2021.

[6] C. Gong, Y. Hu, J. Gao, Y. Wang, and L. Yan, "An improved delay-suppressed sliding-mode observer for sensorless vector-controlled PMSM," *IEEE Trans. Ind. Electron.*, vol. 67, no. 7, pp. 5913–5923, Jul. 2020.

[7] G. Wang et al., "Enhanced position observer using second-order generalized integrator for sensorless interior permanent magnet synchronous motor drives," *IEEE Trans. Energy Convers.*, vol. 29, no. 2, pp. 486–495, Jun. 2014.

[8] N. Ren, L. Fan, and Z. Zhang, "Sensorless PMSM control with sliding mode observer based on sigmoid function," *J. Electr. Eng. Technol.*, vol. 16, no. 2, pp. 933–939, Mar. 2021.

[9] H. Li, Z. Wang, C. Wen, and X. Wang, "Sensorless control of surface-mounted permanent magnet synchronous motor drives using nonlinear optimization," *IEEE Trans. Power Electron.*, vol. 34, no. 9, pp. 8930–8943, Sep. 2019.

[10] G. Mei, W. Lu, Q. Fan, W. Huang, and B. Xiang, "Sensorless PMSM control base on error compensation of sliding mode observer," *Trans. China Electrotechnical Soc.*, vol. 2, no. 38, pp. 398–408, Jan. 2023.

[11] M. Zhang, L. Zhang, X. Zhu, Z. Pei, and K. Xu, "Sensorless control of five-phase flux-intensifying permanent magnet motor based on gradient descent adaptive filter SMO," in *Proc. 26th Int. Conf. Electr. Mach. Syst. (ICEMS)*, Zhuhai, China, Nov. 2023, pp. 491–495.

[12] Z. Wu, C. Cheng, W. Hua, Y. Wang, H. Zhang, and W. Wang, "A frequency-adaptive delay signal cancelation based filter to reduce position estimation error for sensorless IPMSM drives," *IEEE Trans. Power Electron.*, vol. 38, no. 2, pp. 1662–1671, Feb. 2023.

[13] X. Song, J. Fang, B. Han, and S. Zheng, "Adaptive compensation method for high-speed surface PMSM sensorless drives of EMF-based position estimation error," *IEEE Trans. Power Electron.*, vol. 31, no. 2, pp. 1438–1449, Feb. 2016.

[14] Y. Li, W. Li, L. Guo, N. Jin, and F. Lu, "Current sensor-less virtual synchronous generator model predictive control based on sliding mode observer," *IEEE Access*, vol. 9, pp. 17898–17908, 2021.

[15] S. Lin and W. Zhang, "An adaptive sliding-mode observer with a tangent function-based PLL structure for position sensorless PMSM drives," *Int. J. Electr. Power Energy Syst.*, vol. 88, pp. 63–74, Jun. 2017.

[16] T. Wu, X. Wu, S. Huang, K. Lu, and H. Cui, "An optimized PLL with time delay and harmonic suppression for improved position estimation accuracy of PMSM based on Levenberg–Marquardt," *IEEE Trans. Ind. Electron.*, vol. 70, no. 10, pp. 9847–9858, Oct. 2023.

[17] X. Song, J. Fang, and B. Han, "High-precision rotor position detection for high-speed surface PMSM drive based on linear Hall-effect sensors," *IEEE Trans. Power Electron.*, vol. 31, no. 7, pp. 4720–4731, Jul. 2016.

[18] Z. Xu, T. Zhang, Y. Bao, H. Zhang, and C. Gerada, "A nonlinear extended state observer for rotor position and speed estimation for sensorless IPMSM drives," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 733–743, Jan. 2020.

[19] Z. Hao et al., "Phase lead compensation of a phase-locked loop in an IPMSM for air-conditioner compressors," *IEEE Trans. Energy Convers.*, vol. 36, no. 2, pp. 1090–1100, Jun. 2021.

[20] S. Ye and X. Yao, "An enhanced SMO-based permanent-magnet synchronous machine sensorless drive scheme with current measurement error compensation," *IEEE J. Emerg. Sel. Topics Power Electron.*, vol. 9, no. 4, pp. 4407–4419, Aug. 2021.

[21] C. B. Regaya, A. Zaafouri, and A. Chaari, "A new sliding mode speed observer of electric motor drive based on fuzzy-logic," *Acta Polytechnica Hungarica*, vol. 11, no. 3, pp. 219–232, Feb. 2014.

[22] C. Wang and D. Cao, "New sensorless speed control of a hybrid stepper motor based on fuzzy sliding mode observer," *Energies*, vol. 13, no. 18, p. 4939, Sep. 2020.

[23] J. Xu, Z. Wei, and S. Wang, "Active disturbance rejection repetitive control for current harmonic suppression of PMSM," *IEEE Trans. Power Electron.*, vol. 38, no. 11, pp. 14423–14437, Nov. 2023.

[24] R. Sreejith and B. Singh, "Sensorless predictive current control of PMSM EV drive using DSOGI-FLL based sliding mode observer," *IEEE Trans. Ind. Electron.*, vol. 68, no. 7, pp. 5537–5547, Jul. 2021.

[25] S. Golestan, J. M. Guerrero, F. Musavi, and J. C. Vasquez, "Single-phase frequency-locked loops: A comprehensive review," *IEEE Trans. Power Electron.*, vol. 34, no. 12, pp. 11791–11812, Dec. 2019.

[26] C. Yang, J. Wang, X. You, C. Wang, and J. Zhan, "Accuracy comparison of digital implementation on the second-order generalized integrator frequency-locked loop," *Trans. China Electro-Tech. Soc.*, vol. 34, no. 12, pp. 2595–2854, Jun. 2019.

[27] C. Wang, L. Gou, S. Dong, M. Zhou, and X. You, "Sensorless control of IPMSM based on super-twisting sliding mode observer with CVGI considering flying start," *IEEE Trans. Transport. Electrific.*, vol. 8, no. 2, pp. 2106–2117, Jun. 2022.



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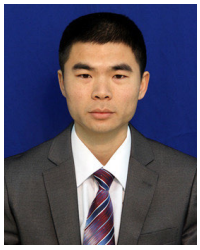
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